

2008 Higher, Section A: scan the code and answer these.



8

10

18

Charged particles, a.c., semiconductors.

Show your working, not just the letter.

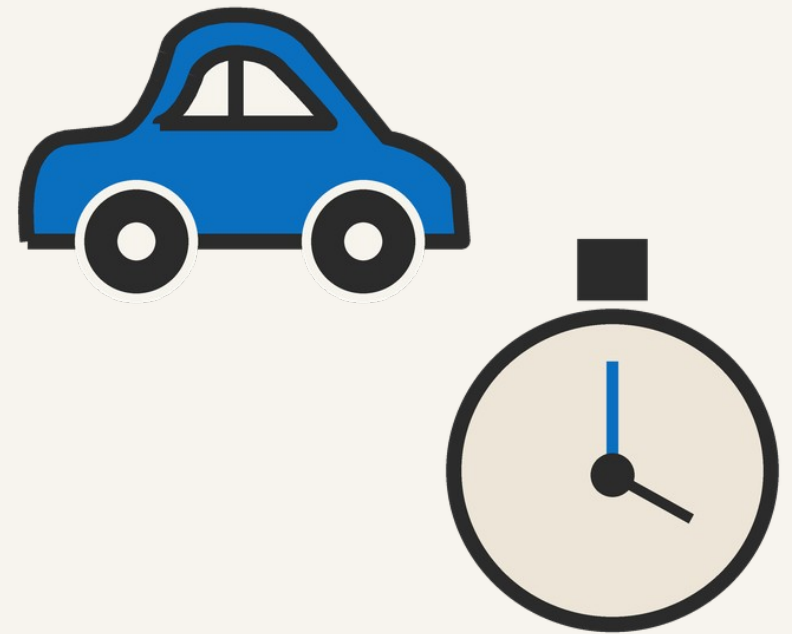


**Scan for the paper
Marking scheme**

Motion, graphs, projectiles

Six lessons on how things move — and
on getting the working right.

Higher Physics



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

PART 1

Motion, as you know it

Everything at Higher is built on the N5 picture of motion.
This part gets it back.

By the end of this part...



- 1 I can tell a scalar from a vector and give examples of each
- 2 I can find distance, displacement, speed and velocity for a journey
- 3 I can find acceleration and displacement from a velocity-time graph

What you already know from National 5

Quantity	Scalar or vector	How you find it
distance, d	scalar	the whole path travelled
displacement, s	vector	start to finish
speed, v	scalar	distance $\div$ time
velocity, v	vector	displacement $\div$ time
acceleration, a	vector	$(v - u) \div t$

Velocity-time graph: gradient is a , area under it is s .

Show Me: motion at National 5. Answer on your board, one at a time.



1

Which of these are vectors: distance, displacement, speed, velocity, acceleration?

velocity, acceleration
and displacement

2

A car drives 300 m north, then 400 m east in 30 s. What distance?

700 m

3

For the same car, what is the size and direction of its displacement?

500 m at 053

Show Me: motion at National 5. Answer on your board, one at a time.



4

For the same car, work out the average speed and the average velocity.

23 m s^{-1} and
 17 m s^{-1} at 053

5

A car goes from rest to 8.0 m s^{-1} in 4.0 s .
Find a , then find s .

$a = 2.0 \text{ m s}^{-2}$
 $s = 16 \text{ m}$

Look back — by the end of this part...



- 1 I can tell a scalar from a vector and give examples of each
- 2 I can find distance, displacement, speed and velocity for a journey
- 3 I can find acceleration and displacement from a velocity-time graph

2011 Higher, Section A: scan the code and answer these.



1

8

18

Scalars and vectors, volts, n-type.

Show your working, not just the letter.



Scan for the paper

No marking scheme

PART 2

The equations of motion

Four equations cover every straight-line problem where the acceleration does not change.

By the end of this part...



- 1 I can pick the right equation of motion from what I am given
- 2 I can solve problems for objects with constant acceleration
- 3 I can lay my working out the way the marking instructions expect

Five symbols, and one condition: constant acceleration

- u — the velocity at the start, in m s^{-1}
- v — the velocity at the end, in m s^{-1}
- a — the acceleration, in m s^{-2}
- s — the displacement, in m
- t — the time taken, in s

Equation 1 – $v = u + at$



$$v = u + at$$

v (m s^{-1})

velocity at the
end

u (m s^{-1})

velocity at the
start

a (m s^{-2})

acceleration

t (s)

time taken

The one with no s in it.

Past-paper example – $v = u + at$

A car and caravan climb a slope at 4.0 m s^{-1} , then speed up uniformly for **250 s** to 9.5 m s^{-1} . Show that $a = 0.022 \text{ m s}^{-2}$.

u v a t s	
u	4.0 m s^{-1}
v	9.5 m s^{-1}
a	?
t	250 s
s	?

$$v = u + at$$

$$a = \frac{v - u}{t}$$

$$a = \frac{9.5 - 4.0}{250}$$

$$\underline{a = 0.022 \text{ m s}^{-2}}$$

Equation 2 — $s = ut + \frac{1}{2}at^2$



$$s = ut + \frac{1}{2}at^2$$

s (m)

displacement

u (m s⁻¹)

velocity at the
start

a (m s⁻²)

acceleration

t (s)

time taken

The one with no v in it.

Past-paper example — $s = ut + \frac{1}{2}at^2$

A car travelling at 12 m s^{-1} accelerates uniformly at -1.5 m s^{-2} for 6.0 s . Calculate the distance travelled in that time.

u v a t s	
u	12 m s^{-1}
v	?
a	-1.5 m s^{-2}
t	6.0 s
s	?

$$s = ut + \frac{1}{2}at^2$$

$$s = (12 \times 6.0) + (\frac{1}{2} \times -1.5 \times 6.0^2)$$

$$s = 72 - 27$$

$$\underline{s = 45 \text{ m}}$$

Equation 3 — $v^2 = u^2 + 2as$



$$v^2 = u^2 + 2as$$

$v \text{ (m s}^{-1}\text{)}$

velocity at the
end

$u \text{ (m s}^{-1}\text{)}$

velocity at the
start

$a \text{ (m s}^{-2}\text{)}$

acceleration

$s \text{ (m)}$

displacement

The one with no t in it.

Past-paper example – $v^2 = u^2 + 2as$

A train speeds up steadily from 5.0 m s^{-1} to 12.0 m s^{-1} while it travels **119 m** along a straight track. Calculate its acceleration.

u v a t s	
u	5.0 m s^{-1}
v	12.0 m s^{-1}
a	?
t	?
s	119 m

$$v^2 = u^2 + 2as$$

$$a = \frac{v^2 - u^2}{2s}$$

$$a = \frac{12.0^2 - 5.0^2}{2 \times 119}$$

$$\underline{a = 0.50 \text{ m s}^{-2}}$$

Equation 4 — $s = \frac{1}{2}(u + v)t$



$$s = \frac{1}{2}(u + v)t$$

s (m)

displacement

u (m s⁻¹)

velocity at the
start

v (m s⁻¹)

velocity at the
end

t (s)

time taken

The one with no a in it.

Past-paper example — $s = \frac{1}{2}(u + v)t$

A car accelerates uniformly from rest and travels **18 m** in 3.0 s.
Calculate the speed of the car at 3.0 s.

u v a t s	
u	0 m s ⁻¹
v	?
a	?
t	3.0 s
s	18 m

$$s = \frac{1}{2}(u + v)t$$

$$18 = \frac{1}{2} \times (0 + v) \times 3.0$$

$$\underline{v = 12 \text{ m s}^{-1}}$$

Choosing the equation: list what you have, and what you want

You have	You want	Use
u, a, t	v	$v = u + at$
u, a, t	s	$s = ut + \frac{1}{2}at^2$
u, a, s	v	$v^2 = u^2 + 2as$
u, v, t	s	$s = \frac{1}{2}(u + v)t$

Worked example – a lorry

A lorry speeds up from 21 m s^{-1} to 35 m s^{-1} over **120 m**.
Calculate its acceleration.

u v a t s	
u	21 m s^{-1}
v	35 m s^{-1}
a	?
t	?
s	120 m

$$v^2 = u^2 + 2as$$

$$a = \frac{v^2 - u^2}{2s}$$

$$a = \frac{35^2 - 21^2}{2 \times 120}$$

$$\underline{a = 3.27 \text{ m s}^{-2}}$$

Worked example – a cannon

A cannon ball starts from rest and accelerates at 150 m s^{-2} for **0.20 s**. Calculate the length of the barrel.

u v a t s	
u	0 m s^{-1}
v	?
a	150 m s^{-2}
t	0.20 s
s	?

$$s = ut + \frac{1}{2}at^2$$

$$s = (0 \times 0.20) + (\frac{1}{2} \times 150 \times 0.20^2)$$

$$\underline{s = 3.0 \text{ m}}$$

Worked example – the same cannon

The same ball accelerates from rest at 150 m s^{-2} for 0.20 s .
Calculate the velocity it leaves the barrel with.

u v a t s	
u	0 m s^{-1}
v	?
a	150 m s^{-2}
t	0.20 s
s	?

$$v = u + at$$

$$v = 0 + (150 \times 0.20)$$

$$\underline{v = 30 \text{ m s}^{-1}}$$

Show Me: the equations of motion. One at a time, on your board.



1

A car speeds up from rest at 2.0 m s^{-2} for 6.0 s . What is its final velocity?

$$v = 12 \text{ m s}^{-1}$$

2

The same car — how far does it travel in those 6.0 s ?

$$s = 36 \text{ m}$$

3

A train slows from 30 m s^{-1} to 12 m s^{-1} over 210 m . Find its acceleration.

$$a = -1.8 \text{ m s}^{-2}$$

4

A cyclist at 4.0 m s^{-1} accelerates at 0.50 m s^{-2} for 8.0 s . How far does she go?

$$s = 48 \text{ m}$$

Show Me: the equations of motion. One at a time, on your board.



5

Which equation would you use if you had u , v and t , and wanted s ?

$$s = \frac{1}{2}(u + v)t$$



Textbook p.28 (1.3.2)

Work through the equations-of-motion questions.
Set every answer out the way the worked examples do.

Look back — by the end of this part...



- 1 I can pick the right equation of motion from what I am given
- 2 I can solve problems for objects with constant acceleration
- 3 I can lay my working out the way the marking instructions expect

2013 Higher, Section A: scan the code and answer these.



3

8

11

Q3 is this experiment – same
apparatus.

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

PART 3

Acceleration down a slope

An experiment the exam can ask you to describe, so it goes in the jotter in full.

By the end of this part...

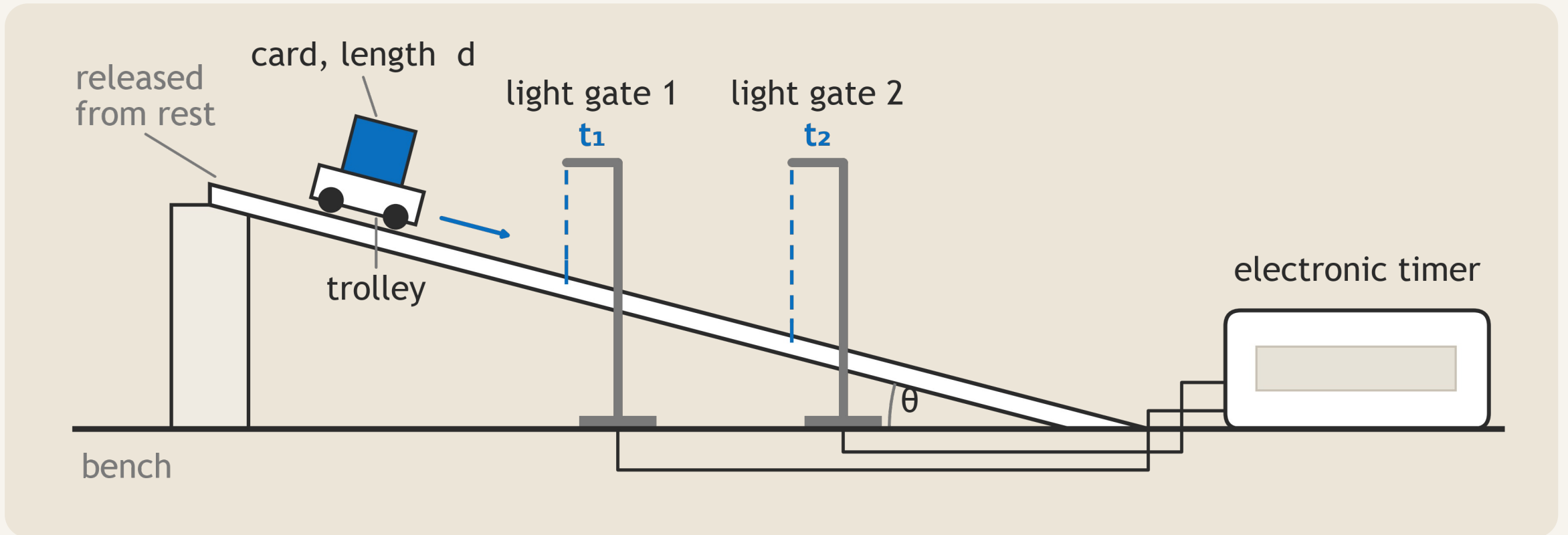


- 1 I can describe an experiment to measure acceleration down a slope
- 2 I can explain why two light gates are needed, not one
- 3 I can calculate the acceleration from the light-gate readings

Why two light gates? Acceleration needs two velocities

- A light gate times how long the card takes to cut the beam.
- Card length $\div$ that time is the velocity at that gate.
- One gate gives one velocity — and no change to measure.
- Two gates give u and v ; the timer gives the t between them.
- Use the same card at both gates, or the lengths differ.

Experiment: acceleration of a trolley down a slope



Draw this, label it, and keep it – the exam can ask you to describe the whole experiment.

The method, step by step



Measure the card length d with a ruler.

Release the trolley from rest, from the same point every run.

Gate 1 times the card cutting its beam: t_1 .

Gate 2 times the same card: t_2 .

The timer also gives t_3 , the time between the gates.

Repeat and take the mean.

Working out the acceleration



$$u = \frac{d}{t_1}$$

$$v = \frac{d}{t_2}$$

$$a = \frac{v - u}{t_3}$$

Worked example – velocity at the first gate

A card **0.025 m** long cuts the first light-gate beam for 0.050 s.
Calculate the velocity of the trolley at that gate.

$$d = 0.025 \text{ m}$$

$$t_1 = 0.050 \text{ s}$$

$$u = ?$$

$$u = \frac{d}{t_1}$$

$$u = \frac{0.025}{0.050}$$

$$\underline{u = 0.50 \text{ m s}^{-1}}$$

Worked example – the acceleration

The card cuts the second beam in 0.020 s, so $v = 1.25 \text{ m s}^{-1}$.

The timer reads 0.35 s between the gates. Find a .

u v a t s	
u	0.50 m s^{-1}
v	1.25 m s^{-1}
a	?
t	0.35 s
s	?

$$a = \frac{v - u}{t_3}$$

$$a = \frac{1.25 - 0.50}{0.35}$$

$$\underline{a = 2.1 \text{ m s}^{-2}}$$

Past-paper example – one gate, and $v^2 = u^2 + 2as$

A trolley is released from rest at P and runs down a ramp.

A card on it cuts a light gate at Q. Which statements are right?

The statement	Right?	Why
$u = 0 \text{ m s}^{-1}$	yes	it starts from rest at P
$s = \text{length of the card}$	no	s is the distance from P to Q
$v = PQ \div \text{timer reading}$	no	the gate times the card, not PQ

Show Me: the slope experiment. One at a time, on your board.



1 What is measured with a ruler in this experiment, and why?

the card length d
gives v at each gate

2 Why is the trolley released from the same point every time?

same u , same distance,
so runs compare

3 Why does the card have to be the same at both gates?

both velocities
use the same d

Show Me: the slope experiment. One at a time, on your board.



4

A card of 0.020 m cuts the beam in 0.040 s. What is the velocity there?

$$v = 0.50 \text{ m s}^{-1}$$

5

Name one thing that makes your value of v too small, and why.

friction – the trolley speeds up less



Textbook p.20 (1.2.4)

Set up the slope, the trolley and the two light gates.
Take three runs, find the mean acceleration, and write the experiment up.

Look back — by the end of this part...



- 1 I can describe an experiment to measure acceleration down a slope
- 2 I can explain why two light gates are needed, not one
- 3 I can calculate the acceleration from the light-gate readings

2015 Higher, Section A: scan the code and answer these.



1

8

10

Vectors, work on a charge, an a.c.
kettle.

Show your working, not just the letter.



Scan for the paper
Marking scheme

PART 4

Straight up and back down

Once it has left your hand, gravity is the only thing acting — so the acceleration never changes.

By the end of this part...



- 1 I can explain why the acceleration is the same all the way up and all the way down
- 2 I can choose a direction as positive and stay with it
- 3 I can solve problems for objects moving vertically

A ball thrown straight up – gravity is the only force

- On the way up it is still moving up, but slowing down.
- At the very top the velocity is zero – the acceleration is not.
- On the way down it speeds up again, at the same rate.
- Slowing down going up is the same as speeding up going down.
- So the acceleration is 9.8 m s^{-2} towards the ground, throughout.

Vertical motion — the sign convention



Choose one direction as positive and keep it for the whole question.

Taking up as positive, $a = -9.8 \text{ m s}^{-2}$ for the whole flight.

The minus sign says the acceleration points down, not that the object does.

At the highest point $v = 0$, but a is still -9.8 m s^{-2} .

A displacement back at the starting height is $s = 0$, not $2h$.

Watch out — signs in vertical motion

THE MISTAKE

- Using $+9.8$ going up and -9.8 coming down
- Writing $s = 0$ as the height reached
- Reading $v = 0$ at the top as $a = 0$

THE CORRECT ANSWER

- One convention for the whole question
- s is the displacement, not the distance flown
- a is -9.8 m s^{-2} at every instant, top included

Markers read every line of your working

IMPORTANT – MARKS

Qualifications Scotland markers check the whole of your working, not only the final line.

A + or – in the wrong place is one of the commonest ways to lose marks here, even when the arithmetic is right.

Write down which direction is positive, then keep every substitution consistent with it.

Worked example – how high?

A stone is thrown straight up, leaving the hand at 10 m s^{-1} .
Calculate the **maximum height** it reaches.

u v a t s	
u	10 m s^{-1}
v	0 m s^{-1}
a	-9.8 m s^{-2}
t	?
s	?

$$v^2 = u^2 + 2as$$

$$s = \frac{v^2 - u^2}{2a}$$

$$s = \frac{0 - 10^2}{2 \times -9.8}$$

$$\underline{s = 5.1 \text{ m}}$$

Worked example – how long?

The same stone leaves the hand at 10 m s^{-1} going straight up.
Calculate the time until it is back at the height it started.

u v a t s	
u	10 m s^{-1}
v	0 m s^{-1}
a	-9.8 m s^{-2}
t	?
s	?

$$v = u + at$$

$$t = \frac{v - u}{a}$$

$$t = \frac{0 - 10}{-9.8}$$

$$t = 1.02 \text{ s to the top}$$

$$\underline{\text{total time} = 2 \times 1.02 = 2.0 \text{ s}}$$

Show Me: vertical motion. One at a time, on your board.



1

A ball is thrown up at 12 m s^{-1} . What is its velocity at the highest point?

$$v = 0 \text{ m s}^{-1}$$

2

What is its acceleration at that highest point?

$$\text{still } -9.8 \text{ m s}^{-2}$$

3

How long does that ball take to reach the top?

$$t = 1.2 \text{ s}$$

4

A stone is dropped from rest down a well and falls for 2.0 s . How fast is it going?

$$v = -19.6 \text{ m s}^{-1}, \text{ so } 20 \text{ m s}^{-1} \text{ down}$$

Show Me: vertical motion. One at a time, on your board.



5

Why is the displacement zero when a ball lands back in your hand?

it finishes
where it started



Textbook p.30 (1.3.3)

Experiment: g by free fall

Vertical-motion questions from the textbook.

Then plan the free-fall experiment: at least five different heights, four repeats each, table and graph in Excel, and a full write-up.

Look back — by the end of this part...



- 1 I can explain why the acceleration is the same all the way up and all the way down
- 2 I can choose a direction as positive and stay with it
- 3 I can solve problems for objects moving vertically

2014 Revised Higher, Section A: scan the code and answer these.



2

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19

Equations of motion, magnetism, p-n.
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

PART 5

Projectiles

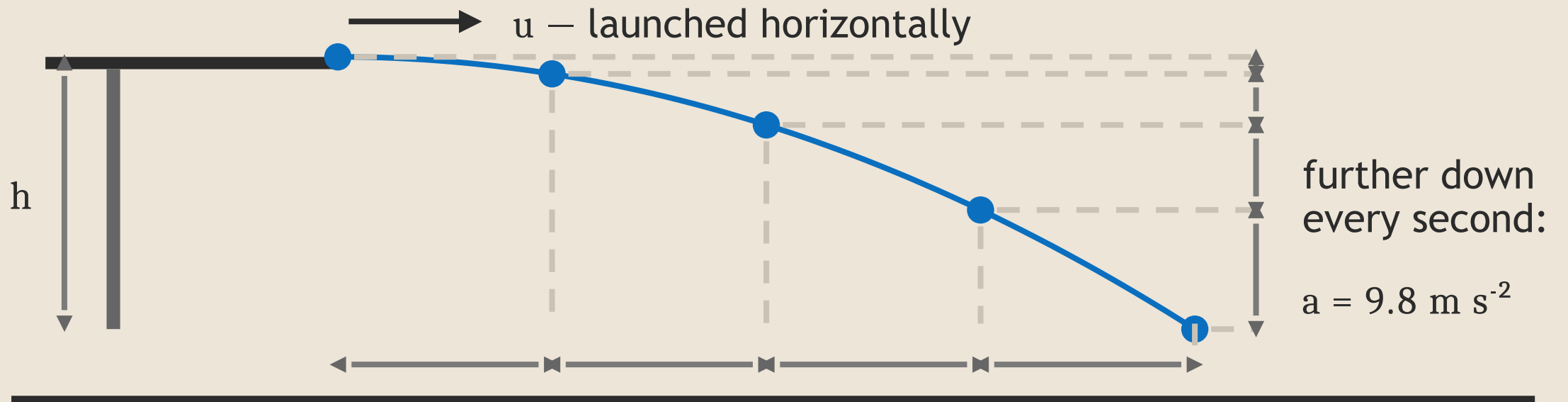
Two motions at once, and they take no notice of each other.

By the end of this part...



- 1 I can explain why horizontal and vertical motion are independent
- 2 I can resolve a launch velocity into its two components
- 3 I can solve problems for a projectile launched at an angle

Projectiles at National 5 – launched horizontally



the same distance across every second – u never changes

Projectiles – the two motions are independent



Horizontal and vertical motion do not affect each other.

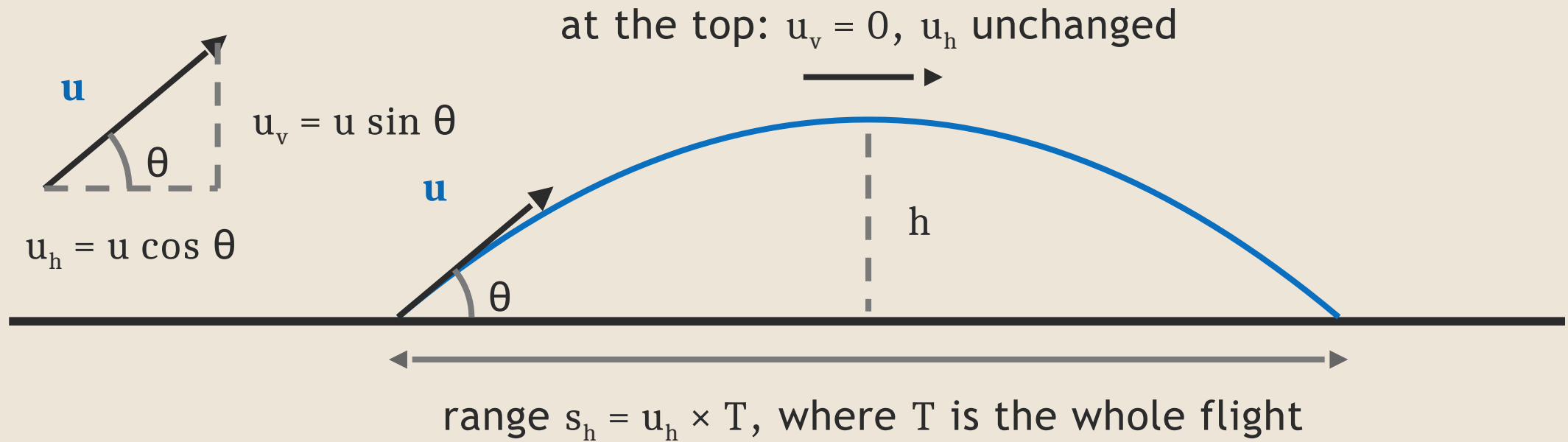
Horizontal: constant velocity, so $s_h = u_h t$.

Vertical: constant acceleration -9.8 m s^{-2} , so use the equations of motion.

Time is the only quantity the two motions share.

A satellite is a projectile that never lands – it is in free fall around the planet.

Higher – launched at an angle



Resolve first: u becomes $u \cos \theta$ across and $u \sin \theta$ up. After that it is an ordinary projectile.

Resolving the launch velocity



$$u_h = u \cos \theta$$

$$u_v = u \sin \theta$$

Then: u_h never changes, and u_v is the u of a vertical problem.

Time of flight is worked out from the vertical motion only.

The u v a t s table – draw this every time



V		H	
u		s_h	
v_v		v_h	
a		t	
t			
s_v			

Setting out a motion problem: the table, every time



Draw the table before you choose an equation.

V is a u v a t s problem — the equations of motion.

H is a constant-speed problem — only $s = v t$.

Fill in what you know; put ? in every row you do not.

The time t is the one row that belongs to BOTH columns.

Worked example – resolving

A ball is launched at 20 m s^{-1} , at 35° to the horizontal.
Calculate the horizontal and vertical components.

V		H	
u	?	s _h	?
v _v	?	v _h	?
a	-9.8 m s^{-2}	t	?
t	?		
s _v	?		

$$u_h = u \cos \theta$$

$$u_h = 20 \cos 35^\circ = 16.4 \text{ m s}^{-1}$$

$$u_v = u \sin \theta$$

$$\underline{u_v = 20 \sin 35^\circ = 11.5 \text{ m s}^{-1}}$$

Worked example – a golf ball (1 of 3)

A golf ball is struck at 30 m s^{-1} , at 25° to the horizontal.

Calculate the horizontal and vertical components of its velocity.

V		H	
u	?	s _h	?
v _v	?	v _h	?
a	-9.8 m s^{-2}	t	?
t	?		
s _v	?		

$$u_h = u \cos \theta$$

$$u_h = 30 \cos 25^\circ = 27.2 \text{ m s}^{-1}$$

$$u_v = u \sin \theta$$

$$\underline{u_v = 30 \sin 25^\circ = 12.7 \text{ m s}^{-1}}$$

Worked example – a golf ball (2 of 3)

The ball leaves the tee with $u_v = 12.7 \text{ m s}^{-1}$ upwards.
Calculate the maximum height it reaches.

V		H	
u	12.7 m s^{-1}	s_h	?
v_v	0 m s^{-1}	v_h	27.2 m s^{-1}
a	-9.8 m s^{-2}	t	?
t	?		
s_v	?		

$$v^2 = u^2 + 2as$$

$$0 = 12.7^2 + (2 \times -9.8 \times s)$$

$$\underline{s = 8.2 \text{ m}}$$

Worked example – a golf ball (3 of 3)

The ball lands at the height it was struck from, so it returns at -12.7 m s^{-1} . Find the time of flight, then the range.

V		H	
u	12.7 m s^{-1}	s_h	?
v_v	-12.7 m s^{-1}	v_h	27.2 m s^{-1}
a	-9.8 m s^{-2}	t	?
t	?		
s_v	?		

$$v = u + at$$

$$-12.7 = 12.7 + (-9.8 \times t)$$

$$t = 2.59 \text{ s}$$

$$\underline{s_h = 27.2 \times 2.59 = 70.4 \text{ m}}$$

Past-paper example – the crossbar challenge (1 of 2)

A football is kicked at 17.0 m s^{-1} , at 24.0° to the horizontal.
Calculate the horizontal and vertical components.

V		H	
u	?	s _h	11 m
v _v	?	v _h	?
a	-9.8 m s^{-2}	t	?
t	?		
s _v	?		

$$u_h = u \cos \theta$$

$$u_h = 17.0 \cos 24.0^\circ = 15.5 \text{ m s}^{-1}$$

$$u_v = u \sin \theta$$

$$\underline{u_v = 17.0 \sin 24.0^\circ = 6.91 \text{ m s}^{-1}}$$

Past-paper example – the crossbar challenge (2 of 2)

The crossbar is 11 m away, and the ball is at the top of its flight as it hits. Find the time to the bar, then the height it hits at.

V		H	
u	6.91 m s ⁻¹	s _h	11 m
v _v	0 m s ⁻¹	v _h	15.5 m s ⁻¹
a	-9.8 m s ⁻²	t	?
t	?		
s _v	?		

$$t = 11 / 15.5 = 0.71 \text{ s}$$

$$v^2 = u^2 + 2as$$

$$s = \frac{0 - 6.91^2}{2 \times -9.8}$$

$$\underline{s = 2.44 \text{ m}}$$

Investigation — which angle goes furthest?

Use the launcher to fire at the same speed
but different angles.

Measure the range each time and put the
results in a table.

How does the launch angle change the
range, and which angle is best?

Write it up in your jotter

Show Me: projectiles. One at a time, on your board.



1

A ball is thrown horizontally off a cliff.
What is its horizontal acceleration?

zero

2

Two identical balls: one dropped, one
thrown horizontally. Which lands first?

together – same
vertical motion

3

A shell is fired at 40 m s^{-1} at 30° . What is
its horizontal component?

$$u_h = 34.6 \text{ m s}^{-1}$$

4

For that shell, what is its vertical
component?

$$u_v = 20 \text{ m s}^{-1}$$

Show Me: projectiles. One at a time, on your board.



5

Why are astronauts on the space station weightless if gravity still acts on them?

they are in free fall
around the Earth



Textbook p.92 (4.2.1)

Projectile questions, including launches at an angle.
Draw the triangle first on every one – components before anything else.

Look back — by the end of this part...



- 1 I can explain why horizontal and vertical motion are independent
- 2 I can resolve a launch velocity into its two components
- 3 I can solve problems for a projectile launched at an angle

2012 Higher, Section A: scan the code and answer these.



8

9

11

Internal resistance, resistors, farads.

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

PART 6

Graphs of motion

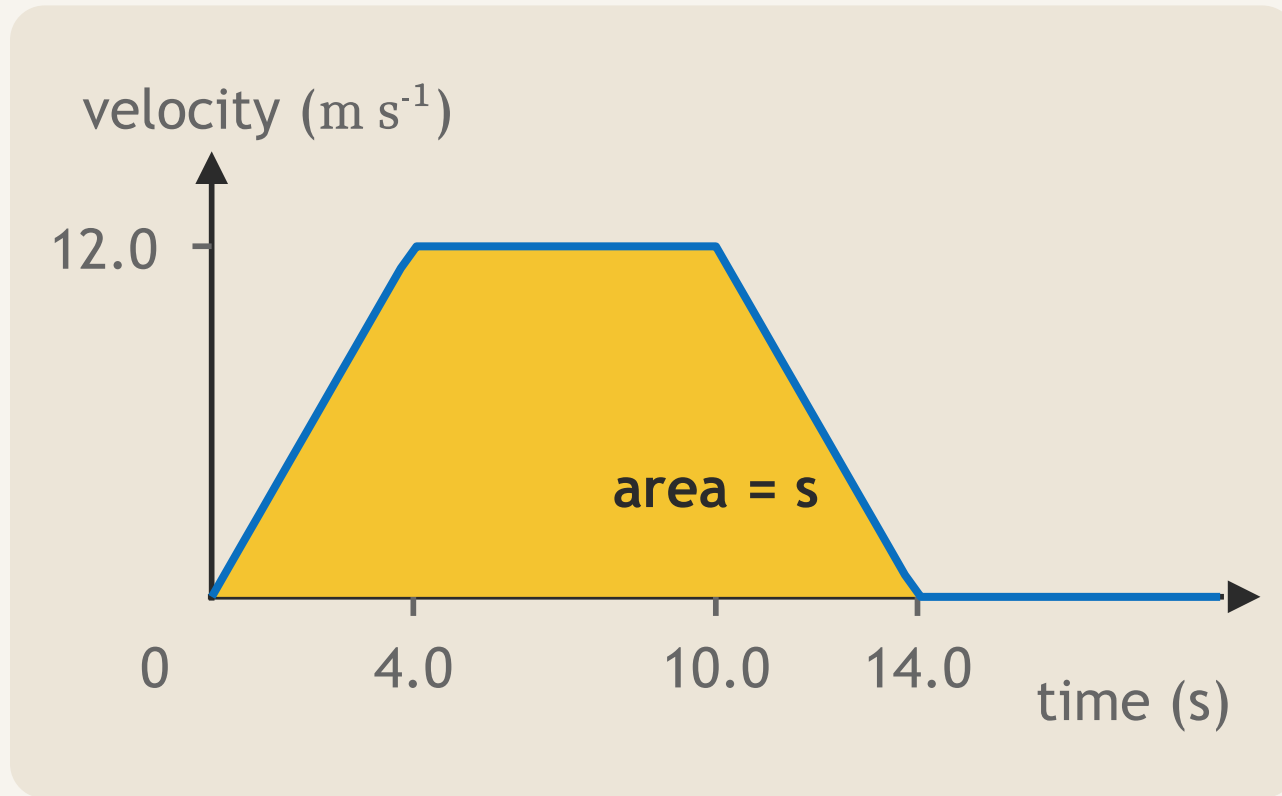
Three graphs of the same journey — and each one tells you what the next one looks like.

By the end of this part...



- 1 I can find velocity, acceleration and displacement from a motion-time graph
- 2 I can change between displacement-time, velocity-time and acceleration-time graphs
- 3 I can sketch the graphs for a ball thrown up and for a bouncing ball

Velocity-time graphs – the gradient is the acceleration



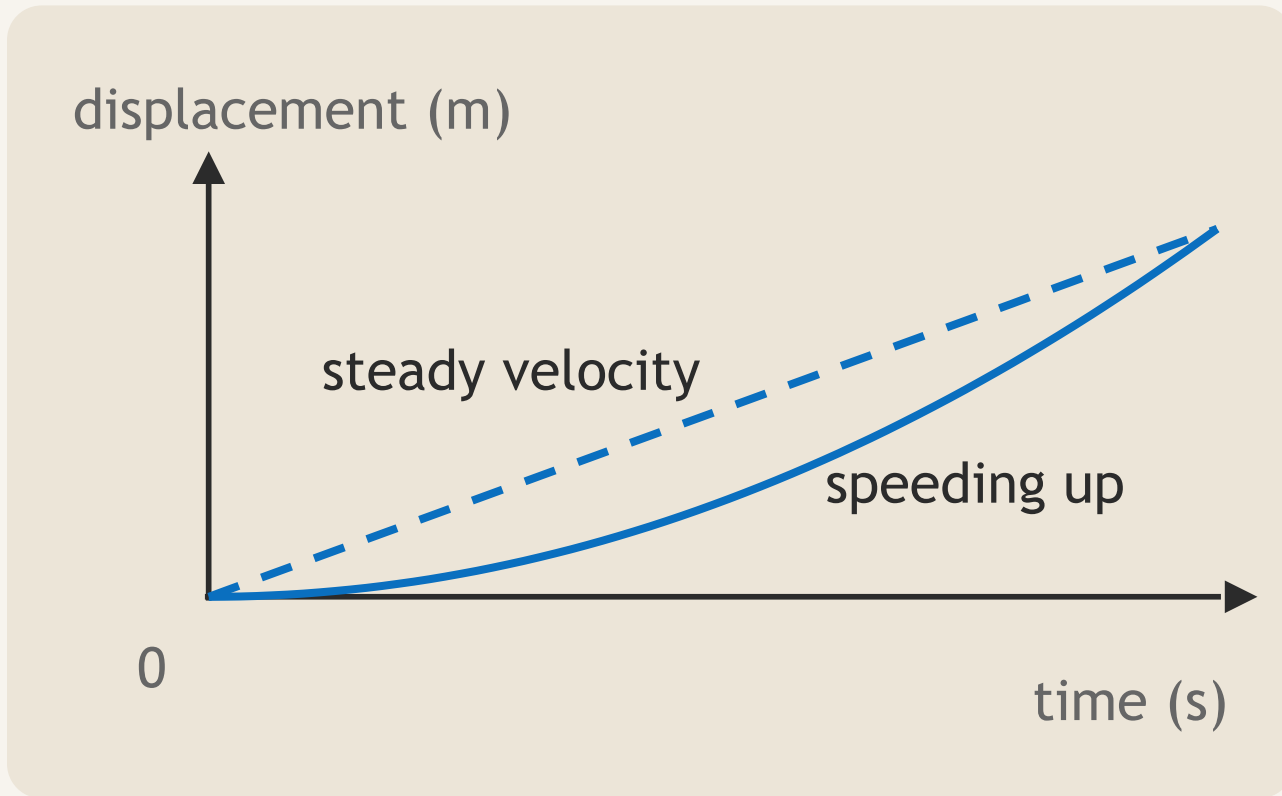
Speeding up: the line rises.

Steady speed: it is flat.

Slowing down: it falls.

The steeper the line, the bigger the acceleration.

Displacement-time graphs – the gradient is the velocity

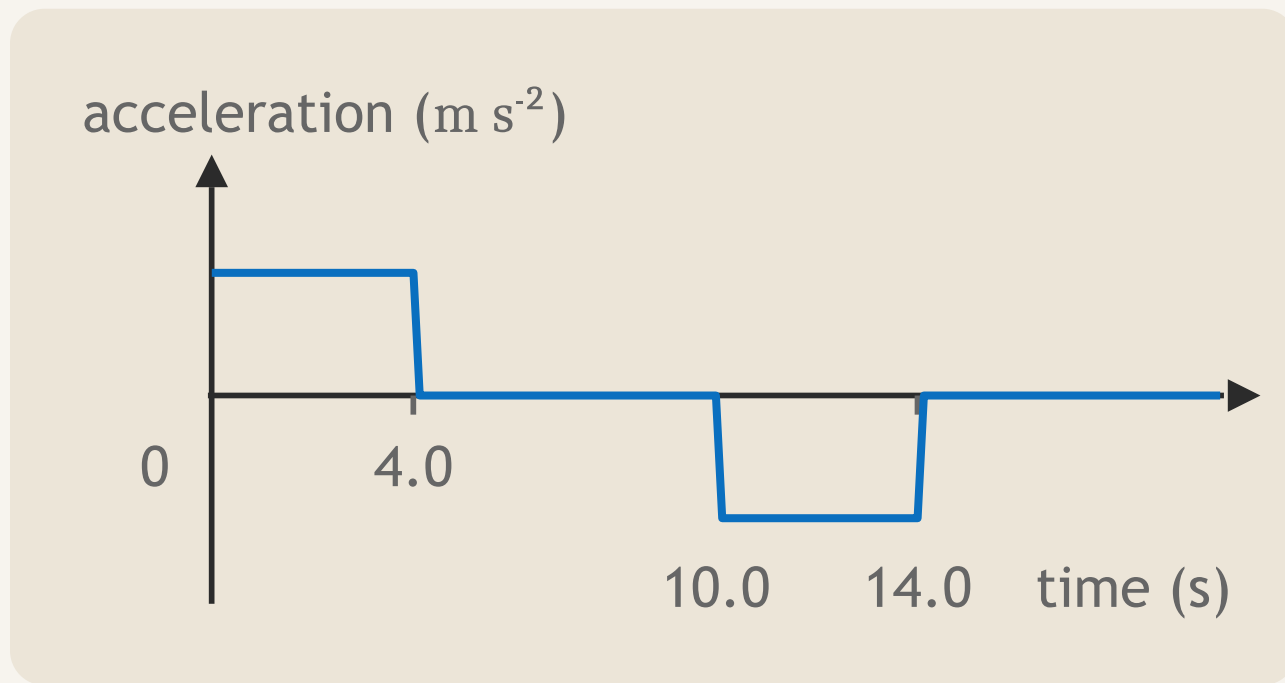


A straight line is a constant velocity.

A curve getting steeper means speeding up.

A flat line means it is not moving.

Acceleration-time graphs – the area is the change in velocity



Constant acceleration is a flat line above the axis.

Steady speed is a flat line on the axis.

Slowing down is a flat line below it.

How the three graphs are related



Gradient of a displacement-time graph = velocity.

Gradient of a velocity-time graph = acceleration.

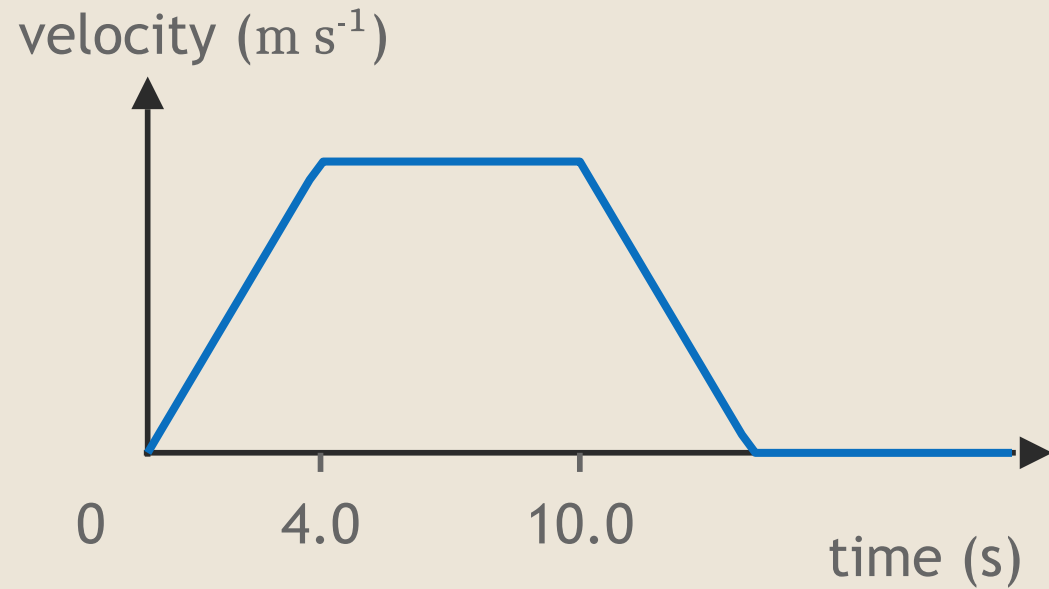
Area under a velocity-time graph = displacement.

Area under an acceleration-time graph = change in velocity.

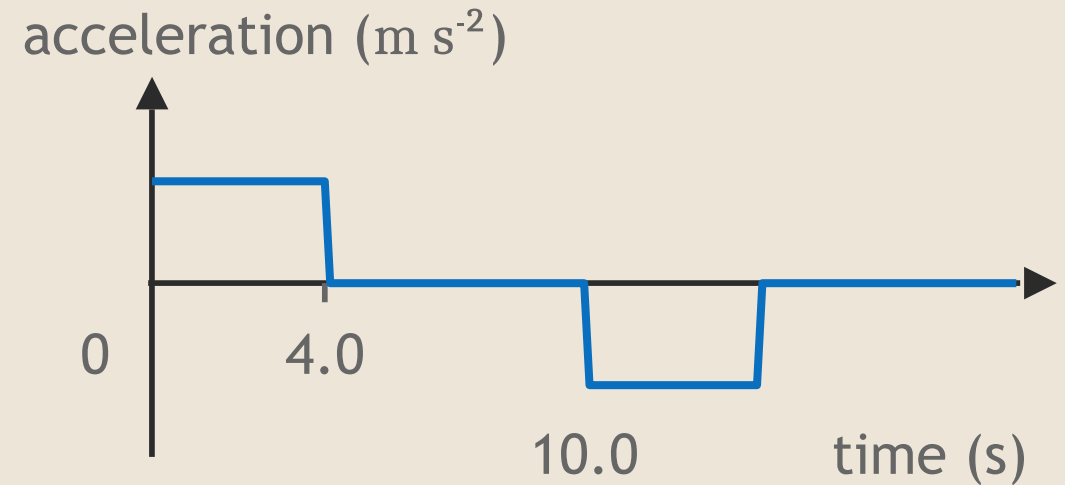
Go down the list with gradients; come back up it with areas.

Changing between graphs – to acceleration-time

velocity-time

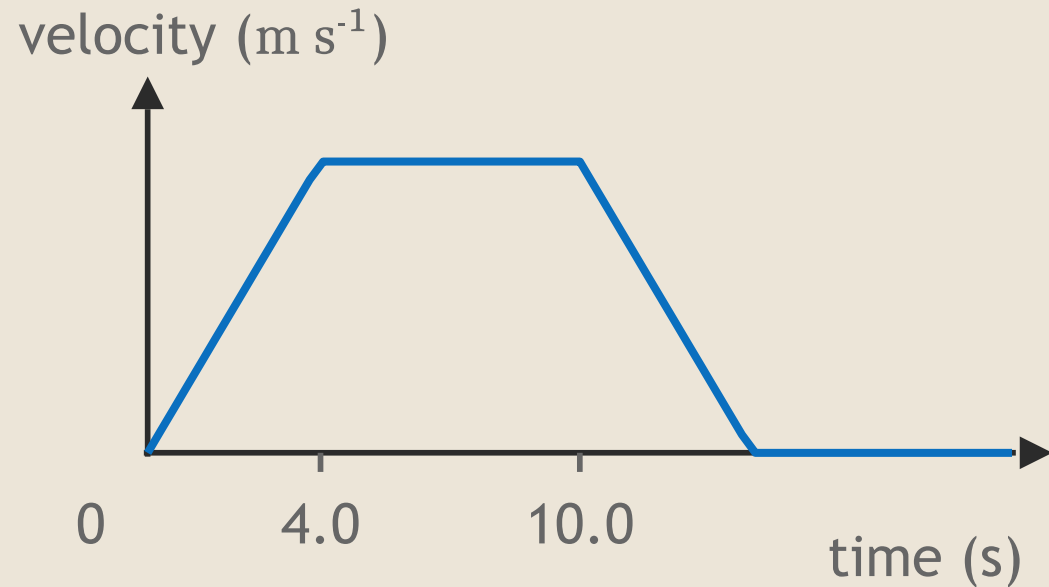


acceleration-time

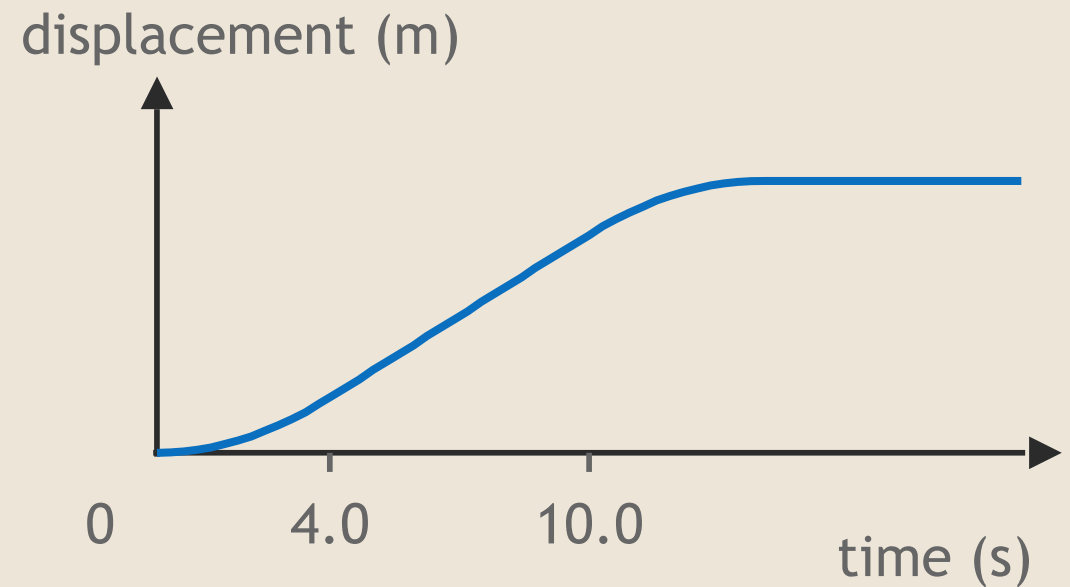


Changing between graphs – to displacement-time

velocity-time



displacement-time



Worked example – reading a velocity-time graph

A car speeds up steadily from rest to 12 m s^{-1} in 4.0 s .
Calculate its acceleration over that 4.0 s .

u v a t s	
u	0 m s^{-1}
v	12 m s^{-1}
a	?
t	4.0 s
s	?

$$v = u + at$$

$$a = \frac{v - u}{t}$$

$$a = \frac{12 - 0}{4.0}$$

$$\underline{a = 3.0 \text{ m s}^{-2}}$$

Worked example – the area under the graph

The same car holds 12 m s^{-1} until 10.0 s , then slows to rest at 14.0 s . Calculate the **total distance** travelled.

speeding up: 0 to 4.0 s

$$\text{area 1} = \frac{1}{2} \times 4.0 \times 12 = 24 \text{ m}$$

steady: 4.0 to 10.0 s

$$\text{area 2} = 6.0 \times 12 = 72 \text{ m}$$

slowing: 10.0 to 14.0 s

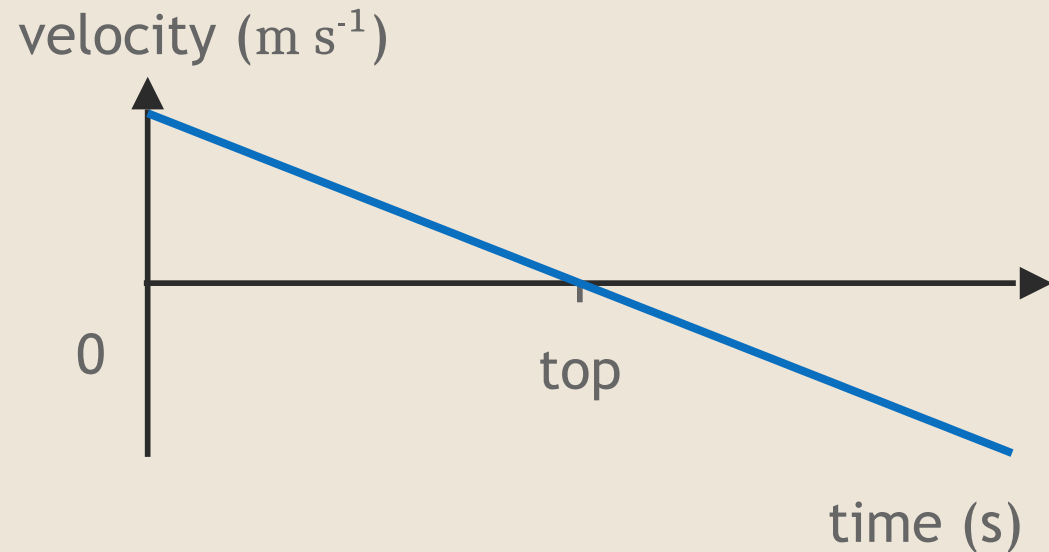
$$\text{area 3} = \frac{1}{2} \times 4.0 \times 12 = 24 \text{ m}$$

distance = ?

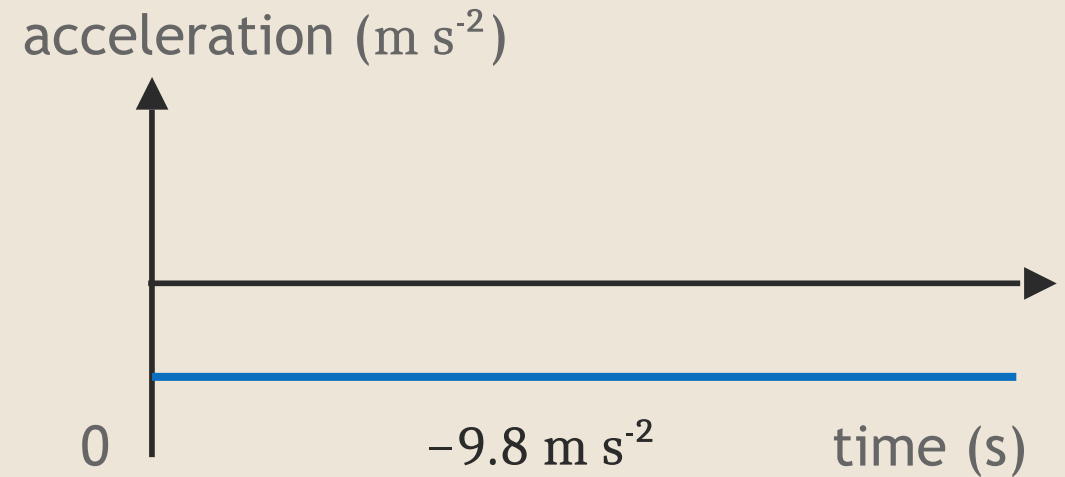
$$\underline{\text{distance} = 24 + 72 + 24 = 120 \text{ m}}$$

A ball thrown straight up, as a velocity-time graph

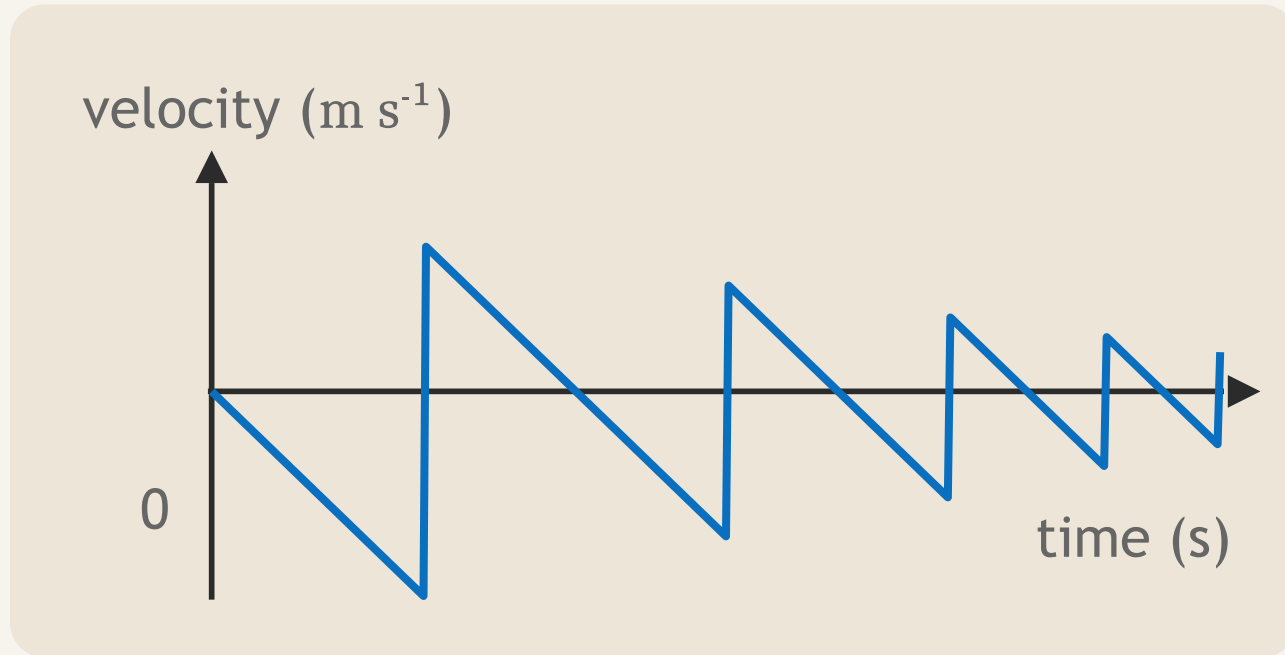
velocity-time



acceleration-time



A bouncing ball, as a velocity-time graph

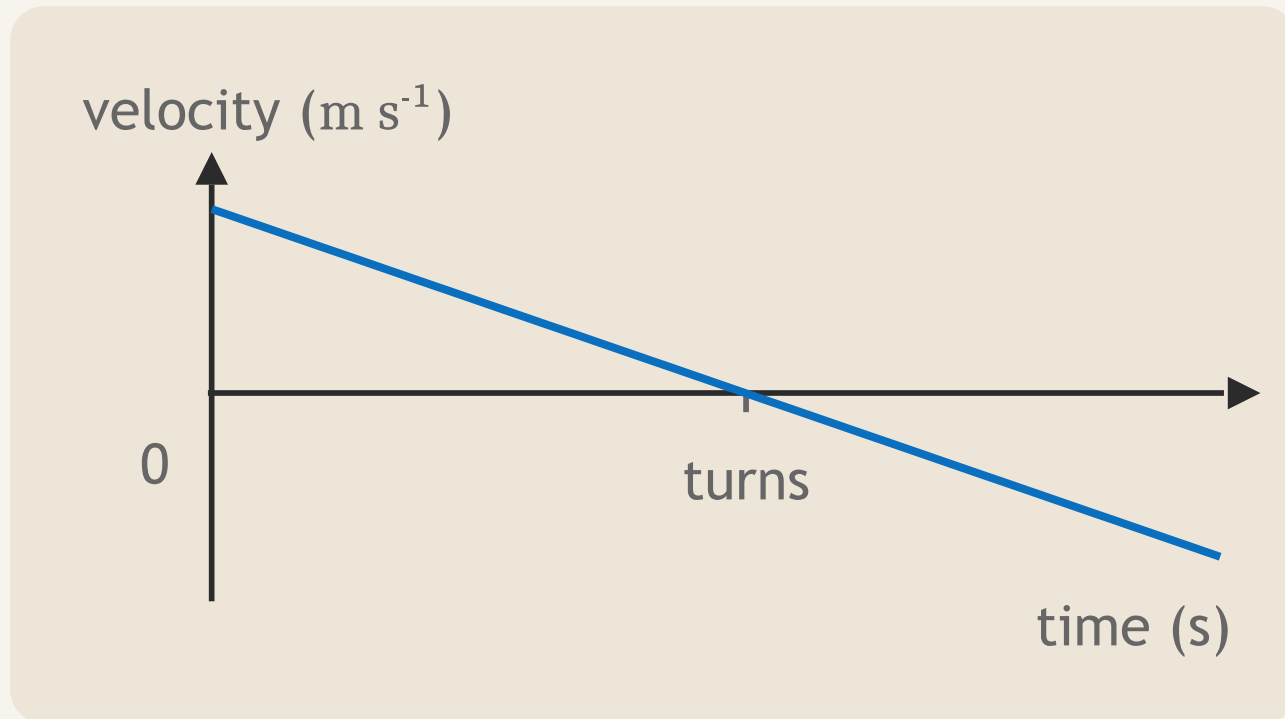


Every sloped line has the same gradient: a is always -9.8 m s^{-2} .

Each jump is a bounce.

The peaks shrink because energy is lost.

A car that changes direction, as a velocity-time graph



Area above the axis is displacement one way.

Area below it is displacement the other way.

Keep the signs for displacement; drop them for distance.

Sketch the graph on your board: axes labelled, units on, origin marked.



1 From rest to 12 m s^{-1} in 4.0 s , then steady for 6.0 s . Sketch v-t.

straight rise to 12,
then flat to 10.0 s

2 Sketch the a-t graph for that same journey.

3.0 m s^{-2} for 4.0 s ,
then zero

3 A ball is dropped and bounces twice. Sketch v-t, down negative.

down, then up smaller
at each bounce

Sketch the graph on your board: axes labelled, units on, origin marked.



4

Sketch the d-t graph of a car parked at the side of the road.

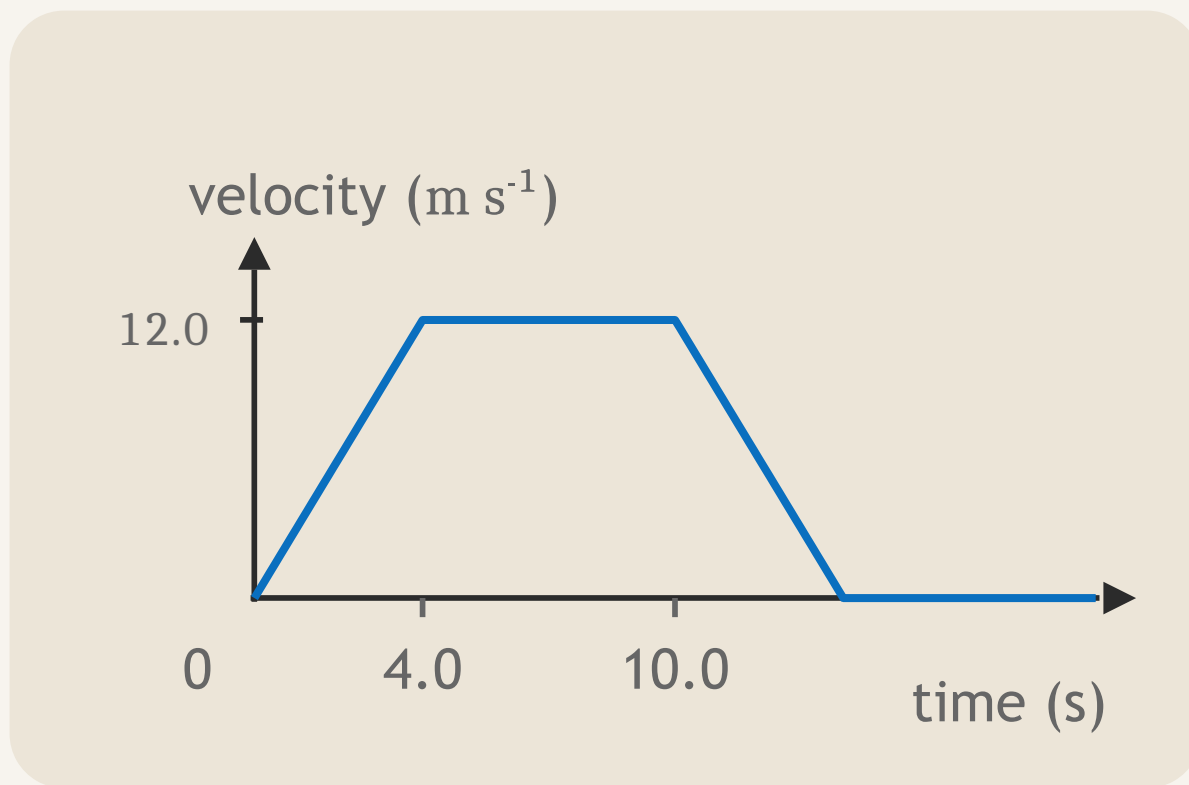
a flat horizontal line

5

Sketch v-t for a car that goes forwards, stops, then reverses.

positive, then zero,
then negative

Read this graph – the journey on the screen



Change this graph. Answer on your board, one at a time.



1 Find the acceleration in the first 4.0 s.

$$a = 3.0 \text{ m s}^{-2}$$

2 Sketch the a-t graph for this journey.

3.0, then zero,
then negative

3 Find the displacement in the first 4.0 s.

$$s = 24 \text{ m}$$

4 Sketch the d-t graph for this journey.

a curve, then
a straight slope

Show Me: graphs of motion. One at a time, on your board.



- 1 What does the gradient of a displacement-time graph tell you?
- 2 What does the area under a velocity-time graph tell you?
- 3 A v-t graph is flat and above the axis.
What is a?

the velocity

the displacement

zero – constant
velocity

Show Me: graphs of motion. One at a time, on your board.



4

Sketch the acceleration-time graph for a ball thrown straight up.

flat at -9.8 m s^{-2}
all the way

5

On a v-t graph, how can you tell it has turned round?

the line crosses
the time axis



Textbook p.17 (Leckie and Leckie)

Graph questions from the textbook.

Sketch every graph in your jotter — axes labelled, with units, and the origin marked.

Look back — by the end of this part...



- 1 I can find velocity, acceleration and displacement from a motion-time graph
- 2 I can change between displacement-time, velocity-time and acceleration-time graphs
- 3 I can sketch the graphs for a ball thrown up and for a bouncing ball

Checkpoint



1

A ball is thrown up and caught. What is its displacement, and why?

zero – it ends where it started

2

Why do the equations of motion fit V but not H ?

vertically a is constant;
horizontally $a = 0$

3

A longer card at the second gate: what happens to a , and why?

v_2 reads low, so a comes out too small

Exam practice – open the paper on your phone

Paper	Question	What it asks
2010 Higher	21	horizontal projectile
2016 Higher	Paper 2, 1	graphs and equations
2003 Higher	21	motion under gravity

LOOKING AHEAD

Forces, energy and power

You can describe how things move. Next: what makes them move that way.

Plenary. Answer on your board, one at a time.



1 Name two vectors and two scalars from this lesson.

displacement, velocity
distance, speed

2 Which equation of motion has no t in it?

$$v^2 = u^2 + 2as$$

3 How would you measure a trolley's acceleration down a slope?

$v = d/t$ at both gates,
then $a = (v - u)/t$

Plenary. Answer on your board, one at a time.



4

A ball is launched at 25 m s^{-1} at 40° .
How would you start?

resolve, then
draw the table

5

What does the area under an
acceleration-time graph give you?

the change in velocity

Extension. Answer on your board, one at a time.



1

Show that 45° gives the greatest range for a fixed launch speed.

range = $u^2 \sin 2\theta / g$,
biggest at $\theta = 45^\circ$

2

Thrown down at 5.0 m s^{-1} from 30 m:
time to land, against dropped?

$t = 2.0 \text{ s}$, against
2.5 s dropped

3

a comes out too small every run. Two reasons, and a test for each.

friction – raise it
card tilted – square it

Course specification – motion, equations and graphs (1 of 3)

- Use of appropriate relationships to solve problems involving distance, displacement, speed, velocity, and acceleration for objects moving with constant acceleration in a straight line.
- Interpretation and drawing of motion-time graphs for motion with constant acceleration in a straight line, including graphs for bouncing objects and objects thrown vertically upwards.
- Knowledge of the interrelationship of displacement-time, velocity-time and acceleration-time graphs.

Course specification – motion, equations and graphs (2 of 3)

- Calculation of distance, displacement, speed, velocity, and acceleration from appropriate graphs (graphs restricted to constant acceleration in one dimension, inclusive of change of direction).
- Description of an experiment to measure the acceleration of an object down a slope.

Course specification – projectiles, taught here (3 of 3)

- Knowledge that the horizontal motion and the vertical motion of a projectile are independent of each other.
- Knowledge that satellites are in free fall around a planet/star.
- Resolution of the initial velocity of a projectile into horizontal and vertical components and their use in calculations.
- Use of resolution of vectors, vector addition, and appropriate relationships to solve problems involving projectiles.

Credits (1 of 3)

2008 Higher, Section A: scan the code and answer these.: © Qualifications Scotland (SQA)

2011 Higher, Section A: scan the code and answer these.: © Qualifications Scotland (SQA)

Past-paper example — $v = u + at$: 2024 Higher, Paper 2 Q2(b) — © Qualifications Scotland (SQA)

Past-paper example — $s = ut + \frac{1}{2}at^2$: 2015 Higher, Section 1 Q2 — © Qualifications Scotland (SQA)

Past-paper example — $v^2 = u^2 + 2as$: 2013 Revised Higher, Section A Q1 — © Qualifications Scotland (SQA)

Past-paper example — $s = \frac{1}{2}(u + v)t$: 2025 Higher, Paper 1 Q1 — © Qualifications Scotland (SQA)

2013 Higher, Section A: scan the code and answer these.: © Qualifications Scotland (SQA)

Past-paper example — one gate, and $v^2 = u^2 + 2as$: 2022 Higher, Paper 1 Q2 — © Qualifications Scotland (SQA)

Credits (2 of 3)

2015 Higher, Section A: scan the code and answer these.: © Qualifications Scotland (SQA)

Watch out – signs in vertical motion: 2024
Higher Physics course report – © Qualifications Scotland (SQA)

2014 Revised Higher, Section A: scan the code and answer these.: © Qualifications Scotland (SQA)

Past-paper example – the crossbar challenge (1 of 2): 2022 Higher, Paper 2 Q1(a)(i) – © Qualifications Scotland (SQA)

Past-paper example – the crossbar challenge (2 of 2): 2022 Higher, Paper 2 Q1(a) – © Qualifications Scotland (SQA)

2012 Higher, Section A: scan the code and answer these.: © Qualifications Scotland (SQA)

Exam practice – open the paper on your phone: © Qualifications Scotland (SQA)

Course specification – motion, equations and graphs (1 of 3): Higher Physics spec v3.0, p.6 – © Qualifications Scotland (SQA)

Credits (3 of 3)

Course specification – motion, equations and graphs (2 of 3): Higher Physics spec v3.0, p.6
– © Qualifications Scotland (SQA)

Course specification – projectiles, taught here (3 of 3): Higher Physics spec v3.0, p.9 –
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